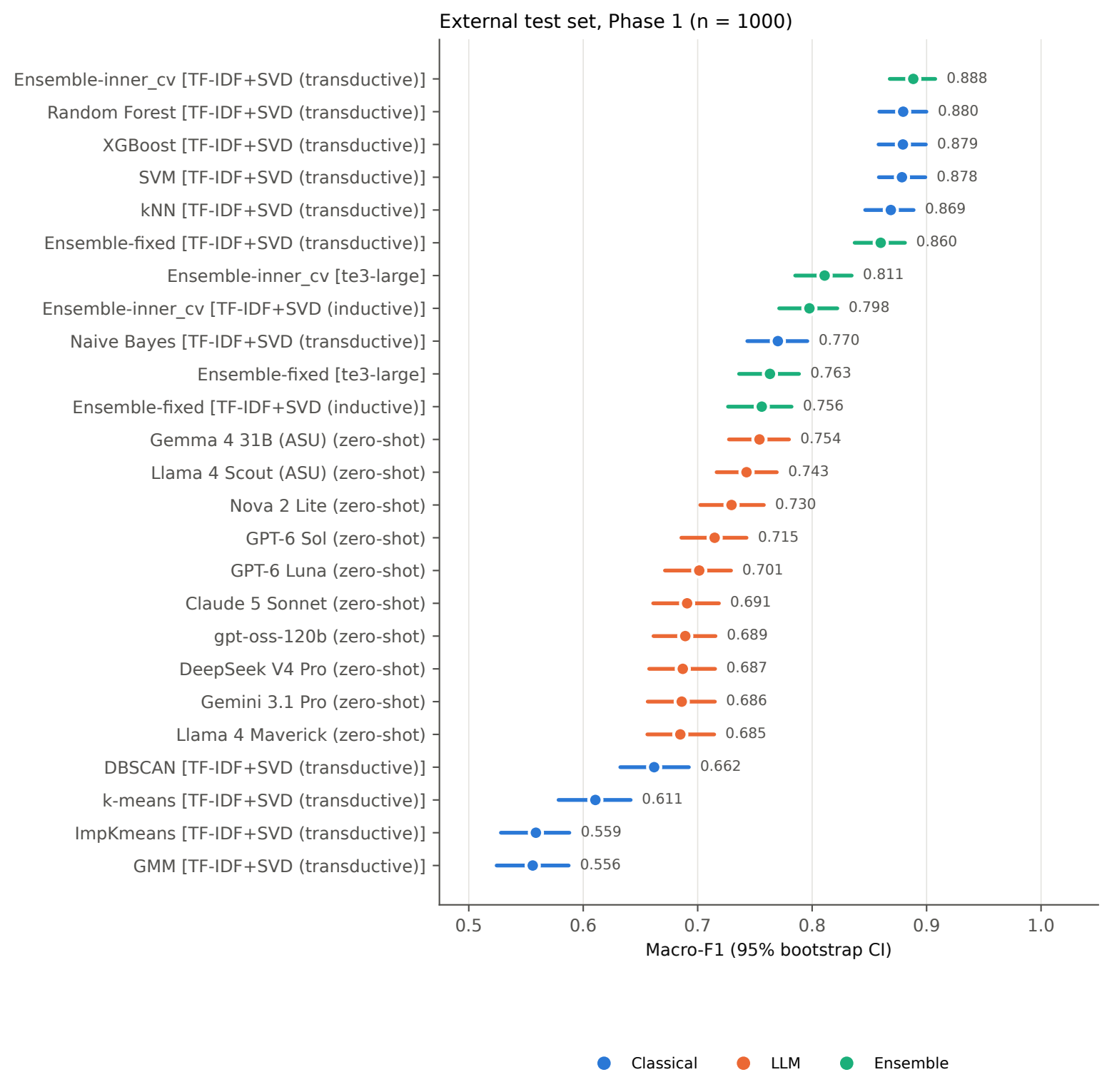
External test set, Phase 1 (n = 1000)
Ensemble-inner_cv [TF-IDF+SVD (transductive)]
Random Forest [TF-IDF+SVD (transductive)]
XGBoost [TF-IDF+SVD (transductive)]
SVM [TF-IDF+SVD (transductive)]
kNN [TF-IDF+SVD (transductive)]
Ensemble-fixed [TF-IDF+SVD (transductive)]
Ensemble-inner_cv [te3-large]
Ensemble-inner_cv [TF-IDF+SVD (inductive)]
Naive Bayes [TF-IDF+SVD (transductive)]
Ensemble-fixed [te3-large]
Ensemble-fixed [TF-IDF+SVD (inductive)]
Gemma 4 31B (ASU) (zero-shot)
Llama 4 Scout (ASU) (zero-shot)
Nova 2 Lite (zero-shot)
GPT-6 Sol (zero-shot)
GPT-6 Luna (zero-shot)
Claude 5 Sonnet (zero-shot)
gpt-oss-120b (zero-shot)
DeepSeek V4 Pro (zero-shot)
Gemini 3.1 Pro (zero-shot)
Llama 4 Maverick (zero-shot)
DBSCAN [TF-IDF+SVD (transductive)]
k-means [TF-IDF+SVD (transductive)]
ImpKmeans [TF-IDF+SVD (transductive)]
GMM [TF-IDF+SVD (transductive)]
0.888
0.880
0.879
0.878
0.869
0.860
0.811
0.798
0.770
0.763
0.756
0.754
0.743
0.730
0.715
0.701
0.691
0.689
0.687
0.686
0.685
0.662
0.611
0.559
0.556
0.5
0.6
0.7
0.8
0.9
1.0
Macro-F1 (95% bootstrap CI)
Classical
LLM
Ensemble